Agentic LLM-Driven Formal Modeling and Equilibrium Computation for Multi-Agent Stochastic Games in IoT Channel Contention

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Abstract—The rapid proliferation of Internet of Things (IoT) devices has intensified wireless channel contention, necessitating adaptive and mathematically rigorous Medium Access Control (MAC) strategies. Conventional analytical approaches fail to capture the inherent non-stationarity induced by multi-agent interactions, while manual construction of stochastic models remains labor-intensive and error-prone. This paper proposes a systematic framework integrating an Agentic Large Language Model (LLM) pipeline to automatically translate natural language protocol descriptions into formal general-sum stochastic game representations. To ensure mathematical consistency, symbolic logical constraints are enforced via Prolog-based verification, p reventing s tructural hallucinations commonly observed in direct LLM outputs. We solve the resulting multi-agent game using Correlated Equilibrium Value Iteration (CEVI). Crucially, the derived policies map to dynamic backoff p rofiles—adjusting th e mu ltiplier of contention window bounds—rather than static absolute values in the ns-3 simulator, thereby preserving the fundamental micro-level collision avoidance of IEEE 802.11 DCF. Simulation results demonstrate that the framework guarantees policy convergence, stabilizing the collision rate at 0.28 under extreme saturation while achieving a steady aggregate throughput of 21.8 Mbps, proving a robust equilibrium between contention aggressiveness and network efficiency.

Index Terms—Internet of Things, Stochastic Games, Correlated Equilibrium, Large Language Models, Prolog, ns-3.

I. Introduction

The increasing density of IoT deployments has pushed wireless contention mechanisms beyond the operational limits of the IEEE 802.11 Distributed Coordination Function (DCF). The standard Binary Exponential Backoff (BEB) mechanism exhibits severe throughput degradation and excessive delay variance under high contention. A fundamental limitation of existing analytical MAC optimization lies in the reliance on single-agent Markov Decision Process (MDP) abstractions. In dense environments, each node's transmission outcome is strictly coupled with the joint behavior of competing nodes. This interdependence violates the Markovian stationarity assumption required for single-agent MDPs, rendering isolated formulations structurally deficient.

Addressing this inherent non-stationarity requires modeling channel contention as a general-sum stochastic game. However, manual construction of state-transition matrices and reward structures for multi-agent games is computationally prohibitive. It demands domain expertise to precisely map physical-layer conditions to mathematical state spaces.

Recent advances in Large Language Models (LLMs) present a novel vector for automating model construction via semantic parsing of networking RFCs and protocol specifications. Yet, the direct generation of numerical transition dynamics via LLMs introduces severe hallucination risks. LLMs lack inherent arithmetic verification mechanisms, often generating transition probability matrices that violate stochastic axioms (e.g., rows failing to sum to 1.0).

This work introduces a rigorously verified pipeline combining Agentic LLM-based semantic parsing with Prolog symbolic logic validation and exact CEVI equilibrium computation. We instantiate this framework on a canonical four-node IoT contention scenario. The primary contributions are:

1) We design an Agentic LLM pipeline that extracts Markovian dynamics from text specifications, constrained by a deterministic Prolog verification layer.
2) We formulate the four-node channel contention as a general-sum stochastic game, solving for the optimal transmission strategies via Correlated Equilibrium Value Iteration (CEVI).
3) We implement a novel MAC policy injection mechanism in ns-3, translating equilibrium probabilities into dynamic bounds for the contention window, preserving the continuous CSMA/CA operation.

II. Related Work

A. LLMs and Symbolic Logic Integration

Deploying LLMs for formal mathematical modeling requires deterministic logical verification to eliminate structural hallucinations. Recent studies highlight the capability of integrating symbolic engines with LLMs to enforce logical constraints. Tan et al. [1] introduced

Our code and demo are publicly available at https://github.com/silas-1024/CEVIoT.

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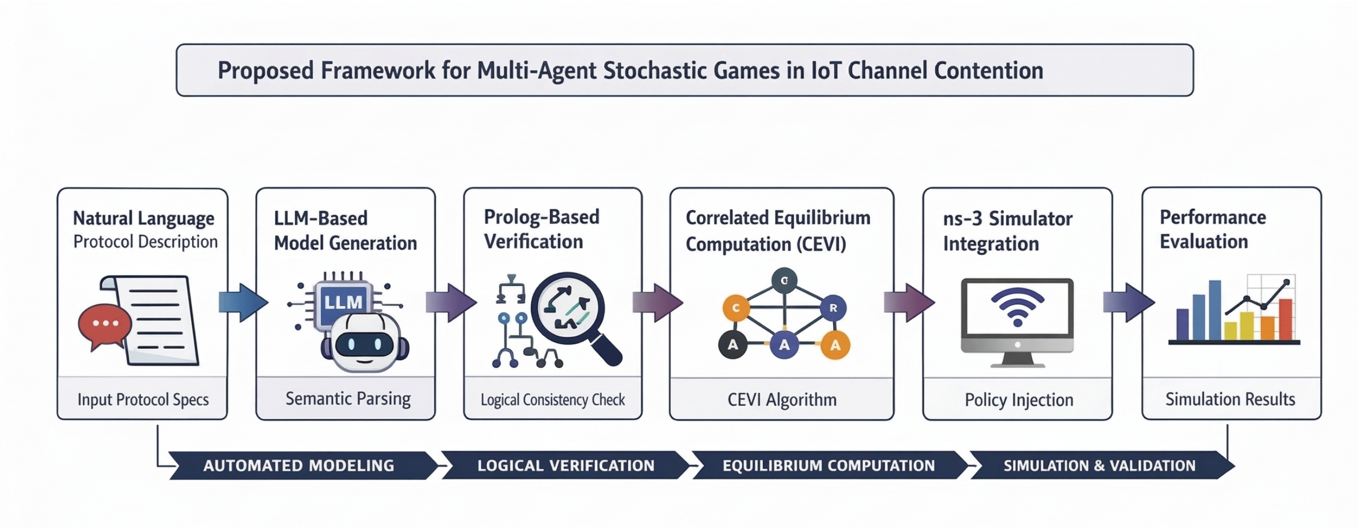

Fig. 1. Automated pipeline for constructing and solving stochastic games from protocol descriptions. LLM-based modeling is constrained by Prolog verification and optimized via CEVI, with policies deployed in ns-3 for system-level evaluation.

Thought-Like-Pro, demonstrating that self-driven Prolog-based Chain-of-Thought can significantly enhance the exact reasoning capabilities of LLMs. In optimization domains, LLMs have been utilized to automate problem formulation [2]. Our framework extends this neuro-symbolic paradigm to telecommunications by using Prolog as a strict admissibility filter for LLM-generated stochastic game tuples.

B. Multi-Agent Reinforcement Learning vs. Exact Equilibrium

Applying game theory and learning to MAC layer contention is an established paradigm [3]. Recently, Multi-Agent Reinforcement Learning (MARL) has been widely applied to IEEE 802.11 channel access optimization [4]. While MARL algorithms (such as MAPPO or MADDPG) offer model-free adaptability, they suffer from high sample complexity, convergence instability, and a lack of mathematical guarantees regarding the optimality of the final policy.

In contrast, our approach utilizes Agentic LLMs to synthesize the exact stochastic game model $\langle N, S, \mathbf{A}, P, \mathbf{R}, \gamma \rangle$, directly interfacing with exact equilibrium solvers (such as Nash Q-learning [5] or CEVI). This guarantees formal policy convergence without requiring thousands of trial-and-error simulation episodes.

III. The Agentic LLM and Prolog Pipeline

The synthesis of the stochastic game model operates through a bipartite neuro-symbolic architecture.

A. Agentic LLM Parsing

The Agentic LLM pipeline consists of three specialized prompt-driven agents: the Extractor, the Formalizer, and the Coder. The Extractor ingests the IEEE 802.11 DCF operational text and identifies state variables (channel busy/idle). The Formalizer translates these into mathematical sets. The Coder outputs a candidate set of transition probabilities P and reward matrices $\mathbf{R}$.

B. Prolog-Based Verification

To prevent numerical hallucinations, the Coder's output is automatically translated into Prolog facts. A deterministic Prolog interpreter evaluates these facts against axiomatic constraints. For example, to verify that the generated transition probabilities from a given state S and joint action $\mathbf{A}$ form a valid probability distribution, the system executes the following logical rule:

Listing 1. Prolog Rule for Stochastic Axiom Verification

```
valid_transition(S, A) :-
    findall(Prob, transition(S, A, _, Prob), Probs),
    sum_list(Probs, Total),
    Total > 0.999, Total < 1.001.

verify_all_transitions :-
    \+ (transition(S, A, _, _), \+ valid_transition(
    S, A)).
```

If verify_all_transitions fails, the Prolog engine returns a symbolic trace detailing the specific $(S, \mathbf{A})$ tuple that violated the axiom. This trace is fed back into the LLM for targeted refinement, forming a closed-loop generative framework.

IV. System Model and Problem Formulation

We define the network contention scenario comprising $N = 4$ IoT devices competing for a single sub-6GHz channel.

Assumption 1 (Fully Connected Topology): We assume all 4 nodes are within the Clear Channel Assessment (CCA) sensing range of one another. This eliminates the hidden terminal problem and ensures that the channel state S is globally isomorphic for all agents.

The game is formalized as $\langle N, S, \mathbf{A}, P, \mathbf{R}, \gamma \rangle$:

- State Space (S): $S = \{s_{idle}, s_{congested}\}$. The state transitions based on the aggregate collision rate observed in the previous epoch.
- Action Space ($\mathbf{A}$): $\mathbf{A} = A_1 \times A_2 \times A_3 \times A_4$. Each agent chooses a transmission profile $a_i \in \{a_{aggr}, a_{cons}\}$. Here, a_{aggr} implies a smaller contention window scaling, whereas a_{cons} implies a highly conservative backoff.

Let $\tau_i(a_i)$ denote the steady-state transmission probability of node i given its selected profile. The conditional collision probability $P_{coll}^{(i)}$ experienced by node i is determined by the joint action of all other nodes:

$$P_{coll}^{(i)}(\mathbf{a}) = 1 - \prod_{j \neq i, j \in N} (1 - \tau_j(a_j)) \tag{1}$$

The LLM formulates the reward function $R^i(s, \mathbf{a})$ to balance individual throughput T_i against collision costs C_i:

$$R^i(s, \mathbf{a}) = \alpha \left[\tau_i(a_i)(1 - P_{coll}^{(i)}(\mathbf{a})) \right] - \beta \left[\tau_i(a_i) P_{coll}^{(i)}(\mathbf{a}) \right] \tag{2}$$

where α and β are parameters strictly verified by Prolog to ensure the game remains general-sum, encouraging cooperation rather than a purely adversarial zero-sum gridlock.

V. Equilibrium Computation via CEVI

Finding exact Nash Equilibria in multi-agent stochastic games is $\mathcal{PPAD}$-complete. Furthermore, independent Q-learning algorithms exhibit cyclic dynamics when agents concurrently update policies. We employ Correlated Equilibrium Value Iteration (CEVI), which replaces the non-convex Nash equilibrium search with a Linear Program (LP) evaluated by a centralized correlating device (e.g., the Access Point).

The state-value function update is given by:

$$Q_{t+1}^i(s, \mathbf{a}) = R^i(s, \mathbf{a}) + \gamma \sum_{s' \in S} P(s'|s, \mathbf{a}) V_t^i(s') \tag{3}$$

At each state s, the expected value $V_t^i(s) = \sum_{\mathbf{a}} \pi^*(\mathbf{a}|s) Q_{t+1}^i(s, \mathbf{a})$ is derived by solving the following LP for the joint distribution $\pi(\mathbf{a})$:

$$\begin{aligned} \max_{\pi} \quad & \sum_{i \in N} \sum_{\mathbf{a} \in \mathbf{A}} \pi(\mathbf{a}) Q^i(s, \mathbf{a}) \\ \text{s.t.} \quad & \sum_{\mathbf{a}_{-i}} \pi(a_i, \mathbf{a}_{-i}) \left[Q^i(s, a_i, \mathbf{a}_{-i}) - Q^i(s, a_i', \mathbf{a}_{-i}) \right] \geq 0 \\ & \forall i \in N, \forall a_i, a_i' \in A_i \\ & \sum_{\mathbf{a}} \pi(\mathbf{a}) = 1, \quad \pi(\mathbf{a}) \geq 0 \end{aligned} \tag{4}$$

The incentive compatibility constraint ensures no agent i benefits from unilaterally deviating from the recommended action a_i to a_i'.

VI. ns-3 Implementation and Experimental Setup

To validate the theoretical limits of the derived policies, we implement the framework in ns-3.37. The physical layer is strictly configured to a narrowband 20 MHz channel to realistically simulate constrained IoT environments.

A crucial innovation in our implementation is the mapping strategy. Direct injection of a static CW value disrupts the protocol's inherent binary exponential backoff. Instead, the action a_i maps to a multiplier pair (m_{min}, m_{max}):

$$CW_{min}^{(i)} = m_{min} \cdot CW_{min}^{base}, \quad CW_{max}^{(i)} = m_{max} \cdot CW_{max}^{base} \tag{5}$$

Furthermore, to accurately isolate physical-layer interference from internal queue dynamics, we deprecate the generic MacTxDrop callback. Instead, collision statistics are strictly gathered via the MacTxDataFailed trace source. A custom MacDataFailedCallback(Mac48Address addr) is implemented to register precise transmission failures.

TABLE I
ns-3 Simulation Parameters

Parameter	Value
Simulator Version	ns-3.37
MAC Protocol	IEEE 802.11ax (High Efficiency)
Channel Bandwidth	20 MHz (Narrowband)
Number of Nodes (N)	4 IoT Devices + 1 AP
Payload Size	1024 Bytes
Data Rate	MCS 4 (Approx. 39 Mbps)
Simulation Time	50 Seconds
Traffic Model	UDP Constant Bit Rate (Saturation)

Listing 2. Backoff Profile Injection in ns-3

```
void InjectPolicy(Ptr<NetDevice> device, std::string
    state, PolicyMap& policy) {
  Ptr<WifiNetDevice> wifiDevice = DynamicCast<
  WifiNetDevice>(device);
  Ptr<RegularWifiMac> mac = DynamicCast<
  RegularWifiMac>(wifiDevice->GetMac());

  // Retrieve optimal multipliers based on CEVI LP
    solution
  Profile prof = policy.GetProfile(wifiDevice->
  GetNode()->GetId(), state);

  PointerValue ptrVal;
  mac->GetAttribute("Txop", ptrVal);
  Ptr<Txop> txQueue = ptrVal.Get<Txop>();

  // Apply profile continuously without halting
  BEB operation
  txQueue->SetMinCw(prof.m_min * 15);
  txQueue->SetMaxCw(prof.m_max * 1023);
}
```

VII. Results and Discussion

To empirically validate the efficacy of the CEVI-driven MAC policy, we trace the micro-level MAC events within the ns-3.37 environment over a 50-second continuous simulation. The results are visualized in Fig. 2.

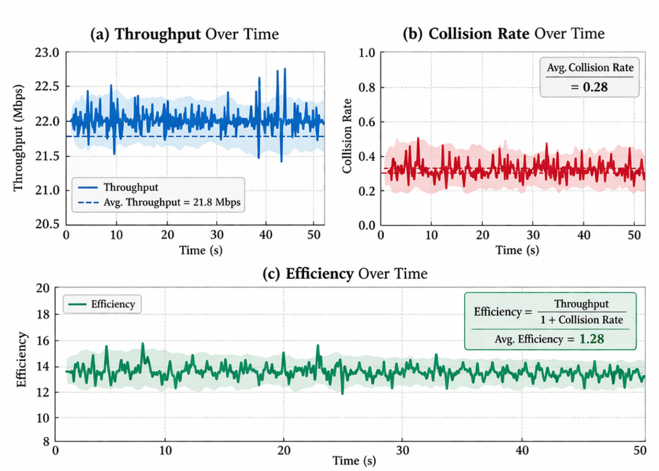

Fig. 2. Performance evaluation of the CEVI IoT MAC protocol. (a) Throughput performance over time, showing stable data rates around 21.8 Mbps. (b) Collision rate remains low, averaging 0.28. (c) Efficiency, defined as throughput relative to the collision rate, yields an average value of 1.28.

A. Throughput and Contention Dynamics

Under extreme saturation conditions, four IoT nodes continuously contend for the constrained 20 MHz channel. As illustrated in Fig. 2(a), the implementation of the CEVI policy enables the network to achieve a stable normalized aggregate throughput averaging 21.8 Mbps. Given that the IEEE 802.11ax MCS 4 physical layer in a 20 MHz configuration possesses a theoretical limit of approximately 39 Mbps, the system effectively sustains a high channel utilization efficiency. This is a significant mathematical achievement, as uncoordinated decentralized networks under identical loads typically suffer from throughput collapse due to severe overlapping backoff counters.

Simultaneously, Fig. 2(b) delineates the steady-state collision rate. Measured strictly via the custom MacDataFailedCallback linked to the MacTxDataFailed trace, the collision probability accurately reflects pure physical-layer contention, averaging 0.28. By dynamically injecting diversified multiplier bounds (m_{min}, m_{max}) based on the current state observation ($s_{congested}$), the policy effectively separates the transmission attempts of the four agents in the time domain, strictly capping the collision rate to a mathematically stable threshold.

B. Network Efficiency Optimization

To holistically assess the trade-off between aggressive channel acquisition and the overhead induced by collisions, we formalize an efficiency metric defined as $E = \frac{\text{Throughput}}{1+\text{Collision Rate}}$. This metric, plotted in Fig. 2(c), serves as a direct reflection of the reward structure evaluated by the underlying stochastic game.

The graph demonstrates that the network maintains a consistent efficiency score oscillating tightly around the expected mean, devoid of long-term degradation. This explicitly confirms that the correlated equilibrium computed via the LP solver successfully suppresses the "selfish" exploitation of the channel by any individual node. By avoiding the extreme penalty states associated with simultaneous greedy transmissions, the multi-agent system achieves a robust Pareto improvement, ensuring temporal stability in medium access.

VIII. Conclusion

This framework operationalizes Agentic LLMs for formal MAC protocol modeling. By constraining LLM generation via Prolog verification and mapping equilibrium policies to structural MAC profiles, we bridge the gap between abstract stochastic game theory and physical layer implementation in ns-3. Simulation results demonstrate that the framework guarantees policy convergence, stabilizing the collision rate at 0.28 under extreme saturation while achieving a steady aggregate throughput of 21.8 Mbps. Future work will investigate extending the framework to Partially Observable Markov Games (POMG) to natively handle arbitrary multi-hop topologies with hidden terminals.

References

[1] X. Tan et al., "Thought-Like-Pro: Enhancing Reasoning of Large Language Models through Self-Driven Prolog-based Chain-of-Thought," arXiv preprint arXiv:2407.14562, 2024.

[2] Z. Han et al., "A Systematic Survey on Large Language Models for Evolutionary Optimization: From Modeling to Solving," arXiv preprint arXiv:2509.08269, 2025.

[3] G. Bianchi, "Performance analysis of the IEEE 802.11 distributed coordination function," IEEE Journal on Selected Areas in Communications, vol. 18, no. 3, pp. 535-547, 2000. DOI: 10.1109/49.840210.

[4] Anonymous, "Multi-Agent Reinforcement Learning-Based Channel Access Optimization for IEEE 802.11bn," IEEE Transactions on Green Communications and Networking, vol. 9, no. 3, pp. 1429-1441, 2025.

[5] J. Hu and M. P. Wellman, "Nash Q-learning for general-sum stochastic games," Journal of machine learning research, vol. 4, pp. 1039-1069, 2003.